

5.2

$f : (a, b) \rightarrow \mathbb{R}$.
 $f'(x) > 0$ on (a, b) .
 $g \equiv f^{-1}$.

To prove: f is strictly increasing in (a, b) .

Proof. Let $x_1, x_2 \in (a, b)$ such that $x_1 < x_2$. We know that f is differentiable on $[x_1, x_2]$. The mean value theorem guarantees the existence of $p \in (x_1, x_2)$ satisfying

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(p).$$

Since $f'(p) > 0$, we have $f(x_2) - f(x_1) > 0$. Thus, f is strictly increasing (a, b) . □

To prove: g is differentiable.

Proof. Let I be the image of f on (a, b) . Let $y \in I$. We have to show that

$$\lim_{t \rightarrow y} \frac{g(y) - g(t)}{y - t} = \frac{1}{f'(g(y))}.$$

Let $x \equiv g(y)$, and $t' \equiv g(t)$. The expression inside the limit then becomes

$$\frac{x - t'}{f(x) - f(t')}.$$

We know that

$$\lim_{t' \rightarrow x} \frac{x - t'}{f(x) - f(t')} = \frac{1}{f'(x)},$$

since $f'(x) > 0$. Let $\epsilon > 0$. There exists $\delta_1 > 0$ such that

$$t' \in (x - \delta_1, x + \delta_1) \implies \left| \frac{x - t'}{f(x) - f(t')} - \frac{1}{f'(x)} \right| < \epsilon.$$

By taking a suitable closed interval $[x - \zeta, x + \zeta]$, we get that g is continuous at y , since continuous bijective maps on compact sets are homeomorphisms. So,

$$\lim_{t \rightarrow y} g(t) = g(y).$$

It follows that there exists $\delta_2 > 0$ such that

$$t \in (y - \delta_2, y + \delta_2) \implies t' \in (x - \delta_1, x + \delta_1).$$

□

5.8

To prove: If f' is continuous on $[a, b]$, f is uniformly differentiable on $[a, b]$.

Proof. WLOG, let $a \leq t < x \leq b$. Since f' is uniformly continuous on $[a, b]$, there exists $\delta > 0$ such that $0 < x - t < \delta \implies |f'(t) - f'(x)| < \epsilon$. Now, from the mean value theorem, there exists $t < t' < x$ such that

$$\frac{f(t) - f(x)}{t - x} = f'(t').$$

So, $0 < x - t < \delta \implies 0 < x - t' < \delta \implies |f'(t') - f'(x)| < \epsilon$. □

To prove: If f' is continuous on $[a, b]$, f is strongly differentiable on $[a, b]$.

Proof. Let $\epsilon > 0$. Consider a sequence $(x_n, y_n) \rightarrow (t, t)$ in $\mathbb{R}^2$. Since f' being continuous on $[a, b]$ implies f is uniformly differentiable on $[a, b]$, there exists $\delta > 0$ such that

$$|x_n - y_n| < \delta \implies \left| \frac{f(x_n) - f(y_n)}{x_n - y_n} - f'(y_n) \right| < \frac{\epsilon}{2}.$$

From slot-wise convergence, we have $(x_n) \rightarrow t$ and $(y_n) \rightarrow t$. Thus, there exists $N_1, N_2 \in \mathbb{Z}^+$ such that $n > N_1 \implies |x_n - t| < \delta/2$ and $n > N_2 \implies |y_n - t| < \delta/2$. From the triangle inequality, we have

$$n > \max\{N_1, N_2\} \implies |x_n - y_n| \leq |x_n - t| + |y_n - t| < \delta.$$

Now, since f' is continuous, there exists $\delta_2 > 0$ such that $|y - t| < \delta_2 \implies |f'(y) - f'(t)| < \epsilon/2$. Also, there exists N_3 such that $n > N_3 \implies |y_n - t| < \delta_2$. Let $N = \max\{N_1, N_2, N_3\}$. Then, we have

$$n > N \implies \left| \frac{f(x_n) - f(y_n)}{x_n - y_n} - f'(t) \right| \leq \left| \frac{f(x_n) - f(y_n)}{x_n - y_n} - f'(y_n) \right| + |f'(y_n) - f'(t)| < \epsilon.$$

Thus,

$$\lim_{(x,y) \rightarrow (t,t)} \frac{f(x) - f(y)}{x - y} = f'(t).$$

□

5.17

f is a real, thrice differentiable function on $[-1, 1]$.
 $f(-1) = 0$, $f(0) = 0$, $f(1) = 1$, $f'(0) = 0$.

To prove: $f'''(x) \geq 3$ for some $x \in (-1, 1)$.

Proof. On applying Taylor's theorem with $\alpha = 0$ and $\beta = 1$, we get

$$\begin{aligned} f(1) &= \frac{f''(0)}{2} + \frac{f'''(t)}{6} \\ \implies 6 &= 3f''(0) + f'''(t). \end{aligned}$$

for some $t \in (0, 1)$. With $\beta = -1$, we get

$$\begin{aligned} f(-1) &= \frac{f''(0)}{2} - \frac{f'''(s)}{6} \\ \implies f'''(s) &= 3f''(0). \end{aligned}$$

for some $s \in (-1, 0)$. Thus, we have

$$f'''(s) + f'''(t) = 6.$$

This is impossible if $f'''(x) < 3$ for all $x \in (-1, 1)$. □